

IT643 - Software Design and Testing

Course Placement: Autumn 2025 – Core MSc IT

Course Format: It is 3 lectures and one lab every week

Prerequisites (if any): Awareness of Object oriented programming

Credit Structure (L-T-P-Cr): 3-0-2-4

Instructor's Name with email: Ankush Chander (ankush_chander@dau.ac.in)

Course Objective:

Course aims to impart students know-how of creating modular, flexible, and reusable software, by applying object-oriented design principles and guidelines. And, you will be able to communicate these designs in a visual notation known as Unified Modelling Language (UML).

Course Content:

- Evolution of programming- Programming Paradigms
- Software design vs Software Architecture
- Role of design in software process
- Object Oriented principles: Classes (Data + behaviour), Categories of objects, Encapsulation, Abstraction, Generalization, Decomposition
- Modelling using CRC(Class Responsibilities Collaborators) cards and UML diagrams
- Design principles - Coupling and Cohesion, Separation of Concerns, Information Hiding, Conceptual Integrity
- Software Testing: Basic concepts and preliminaries,
- Design and Code Inspections to Reduce Errors in Program Development,
- Functional Testing (Black Box, White Box, Mutation Regression)
- Debugging,
- Verification of Software Systems, Testing of Web GUI and Mobile Apps
- Non-functional Testing.

Suggested Textbook/references:

1. [Head First Object-Oriented Analysis and Design](#)
2. [Software Design and Architecture Specialization](#)

Assessment Method:

- Lab +Group Project assignments: 50% (group size will be a maximum of 4 students)
- Mid-Sem Exam: 25%
- End Sem Exam: 25%

Grading Policy: Relative grading with the following slabs:

- >90 = AA
- 80-90 = AB
- 65-79 = BB
- 50-64 = BC
- 40-50 = CC
- <39 = CD

Course outcome:

Upon completion, students will be equipped with the advanced principles, modern techniques, and practical skills required to architect software that is inherently testable, maintainable, reflecting the core tenets of creating software that is "safe from bugs, easy to understand, and ready for change".